A web system for traceable exploration and analysis of heterogeneous earthquake catalogues

José Manuel Villa Vargas; Gabriel Hurtado Avilés[1]*

[1] Escuela Superior de Cómputo (ESCOM), Instituto Politécnico Nacional, Mexico City, Mexico

*Corresponding author: Gabriel Hurtado Avilés. Email: ghurtadoa@ipn.mx.

ORCID: José Manuel Villa Vargas, https://orcid.org/0009-0009-2401-0661; Gabriel Hurtado Avilés, https://orcid.org/0009-0002-5686-1822.

Abstract

Earthquake catalogues are distributed across agencies and differ in coverage, fields and parameter versions. These differences complicate joint exploration and the reuse of derived information. This study develops and evaluates a web system that supports catalogue consultation, comparison and descriptive analysis while preserving provenance and interpretation limits. The contribution comprises the computational system, documented seismic inputs and reproducible results that demonstrate how the system addresses this problem. A static browser application coordinates maps, filters, annual record counts and source inspection; CSV exports and hash-linked query specifications retain the selection for subsequent analysis, study and research. Four frozen catalogues distributed by USGS, NRCan and SSN supply 287,936 source records. Separate and combined views preserve their different coverage, missing fields and shared identifiers rather than asserting a unified population of independent earthquakes. Demonstrations include a direct SSN selection of 204 records in partial 2026 with reported magnitude at least 4.5, and a combined selection of 7,107 records for 2010-2025 at the same reported-magnitude threshold. Reproducible offline pipelines extend browser exploration with annual SSN summaries and retained-revision comparisons. A 36-event ComCat pilot contains 236 public origin-product versions; 13 of 31 same-type endpoint pairs change magnitude and four cross 5.5. Independent SQL evaluation agrees with 4,720 conditional retained projections. Explicit evidence contracts distinguish captured observations, calculations on retained versions and unsupported historical reconstructions. The resulting workflow produces traceable selections and descriptive information for further investigation without homogenizing magnitudes or inferring completeness, historical public states or earthquake forecasts. Source accuracy, historical reconstruction accuracy and usability with independent users remain unevaluated.

Keywords: earthquake catalogues; geovisualization; provenance; retained revisions; reproducibility; evidence contracts

1. Introduction

Earthquake catalogues differ in geographic windows, observation periods, measurement fields and revision practices. Researchers must account for these differences when selecting records and reproducing derived results. Established catalogue preparation addresses source heterogeneity, duplicates and magnitude compatibility (Weatherill et al., 2016); Mexican applications also require declustering and completeness assessment (Sawires et al., 2019). Aggregation alone does not perform these scientific operations.

The problem addressed here is how to make heterogeneous earthquake records jointly explorable and reusable without losing the distinctions needed to interpret them. The objective is to develop and evaluate a web system for consulting, comparing and exploring seismic catalogues, and for obtaining traceable descriptive information that supports analysis, study and subsequent research. Users should be able to define a selection, inspect its source records, derive summaries and retain the inputs and query needed for further work. Derived information must remain distinguishable from agency-reported values.

The article presents three connected contributions: the computational system with coordinated views, source inspection and reproducible exports; documented seismic inputs and transformations; and results demonstrating catalogue selection, SSN descriptive summaries and retained-revision analysis. These tasks address coverage, source attribution and parameter stability. They demonstrate functional capabilities rather than measured gains in productivity.

Temporal interpretation requires particular care. ComCat exposes versioned products and preferred summaries (U.S. Geological Survey, n.d.a, n.d.b, n.d.c), whose evolution has been studied by Thompson et al. (2020). Libcomcat, Timeline and INGV provide history capabilities (U.S. Geological Survey, n.d.d, n.d.e; Pintore, 2026). Building on those tools and resolution-aware query research (Sismanis et al., 2009; Gardezi and Bertossi, 2011), executable evidence contracts specify which derived calculations an externally acquired archive supports when historical completeness is unknown.

Following Villa Vargas et al. (2026), this work uses separately documented inputs and a newly acquired revision cohort. Figure 1 introduces catalogue selection, inspection and reuse. The system produces no earthquake predictions or damage estimates.

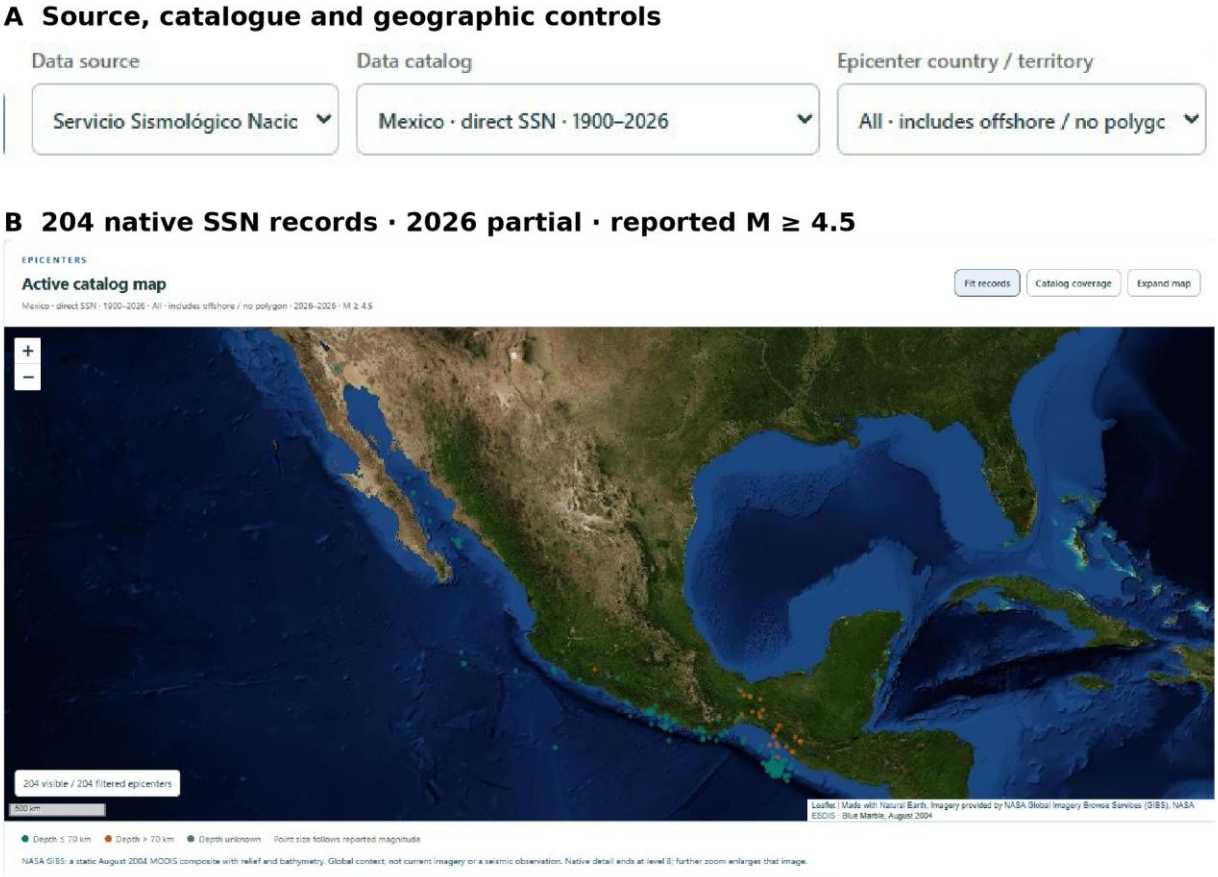


Figure 1. Catalogue exploration in the system. (A) Enlarged source and catalogue controls. (B) The same 204 records selected from the direct SSN download for origin year 2026 and reported magnitude at least 4.5, through 6 November (partial year). The English interface uses the blue palette and light appearance. NASA Blue Marble's static August 2004 composite supplies background context; NASA GIBS/ESDIS credits remain visible. Both frames come from one unchanged Edge capture. The selection supports CSV and local-query JSON exports; the map demonstrates functionality, not source accuracy, earthquake probability or historical public-state reconstruction.

2. Computational system and analytical methods

2.1. System architecture and browser workflow

The computational workflow separates acquisition, preserved raw evidence, reproducible data preparation and static browser delivery. Catalogue builders produce source-specific packages and manifests; the browser consumes these prepared assets without requiring a database server. Reproducible Python pipelines perform the SSN annual-maximum and retained-revision analyses offline; the revision view presents prepared results. Interactive browser

operations provide filtered records, annual counts, inspection and exports for downstream work, rather than a general statistical modelling environment.

The static application coordinates a map, annual record counts and table over one catalogue or the four-package composition. Coordinated views follow established practice (Wu et al., 2020), without claiming a new clustering method. Package bytes are hash-checked before export. Data source and background remain separate. Five backgrounds comprise local Natural Earth, OpenStreetMap, humanitarian OSM, USGS Topo and NASA Blue Marble's static August 2004 composite; external tiles load on demand with attribution.

Country/territory filtering uses reported coordinates within 47 generalized Natural Earth 1:50 million land regions (Natural Earth, n.d.). Point-in-polygon evaluation preserves holes and antimeridian coordinates. Edges or overlapping regions yield boundary; no land containment yields outside, not verified marine membership; unusable coordinates yield invalid. It infers neither nationality nor maritime jurisdiction, exclusive economic zones, nearest country or locality-based membership. A sidecar, SHA-256 9063e0c34412fec41debad30f69a653486c18e0f94d5983e6a061847befe27cb, aligns one label to each package row; event hashes and original row order bind the classification to unchanged inputs.

Filters cover inclusive UTC origin-year and reported-magnitude bounds, exact magnitude-type/agency strings, depth in documented kilometres, missing horizontal error, and ID/place text. Accent-insensitive search preserves original text. Canadian depth values remain outside the kilometre field. The separate SSN/UNAM-via-ComCat shortcut selects Mexico-package rows whose locationSource or magSource equals unm or us_unm, preserving USGS distribution. Native SSN and the 36-event revision pilot remain separate.

Composition retains all 287,936 source rows in registry order, using catalogId~packagedId keys and preserving each original key, identifier kind, provider, snapshot and fields. The 2,163 ComCat IDs shared by the two USGS packages are flagged without merging. No fuzzy, geographic or temporal cross-provider matching is performed; counts describe records, not unique or independent earthquakes.

CSV preserves source fields and adds provenance/derived geography. JSON seismic-local-query/4 documents country-filtered or combined queries, including each input's hashes, geographic index/geometry hashes, predicates and ordering. Combined exports retain source-qualified result keys. Corresponding v4 URLs retain the geography hash. Single-catalogue views without a country filter retain v3; v1/v2/v3 links remain readable, with v1 selecting the original Mexico snapshot. These are local operations, not remote requests or historical replays.

Three palettes and system/light/dark appearance remain separate from query state. Removable filters update coordinated views. Table-to-map focus, expanded map, return button and Escape support inspection (Figure 2); navigation preserves filters and reported measurements.



Figure 2. Combined expanded-map view: UTC origin years 2010-2025, reported magnitude at least 4.5, all country/territory classes including outside and boundary. (A) Enlarged return and background controls. (B) Actual map reporting 7,073 visible of 7,107 selected records; 34 lie outside the frame. Both panels derive from one unchanged local Edge capture of the released system. Contributions are 2,114 North American USGS, 2,026 Mexico-region USGS, 377 NRCan and 2,590 SSN records. Map extent does not filter CSV/JSON exports, and input coverage differs. The Spanish interface uses maroon/light appearance over NASA's static August 2004 Blue Marble composite, with NASA GIBS/ESDIS credits. This selection differs from Figure 1; counts denote source records, not unique or independent earthquakes.

The application has two connected analytical paths: fixed-catalogue exploration and the retained-revision audit (Figure 3). Both retain source roles and hashes; their counting units and evidential claims remain distinct.

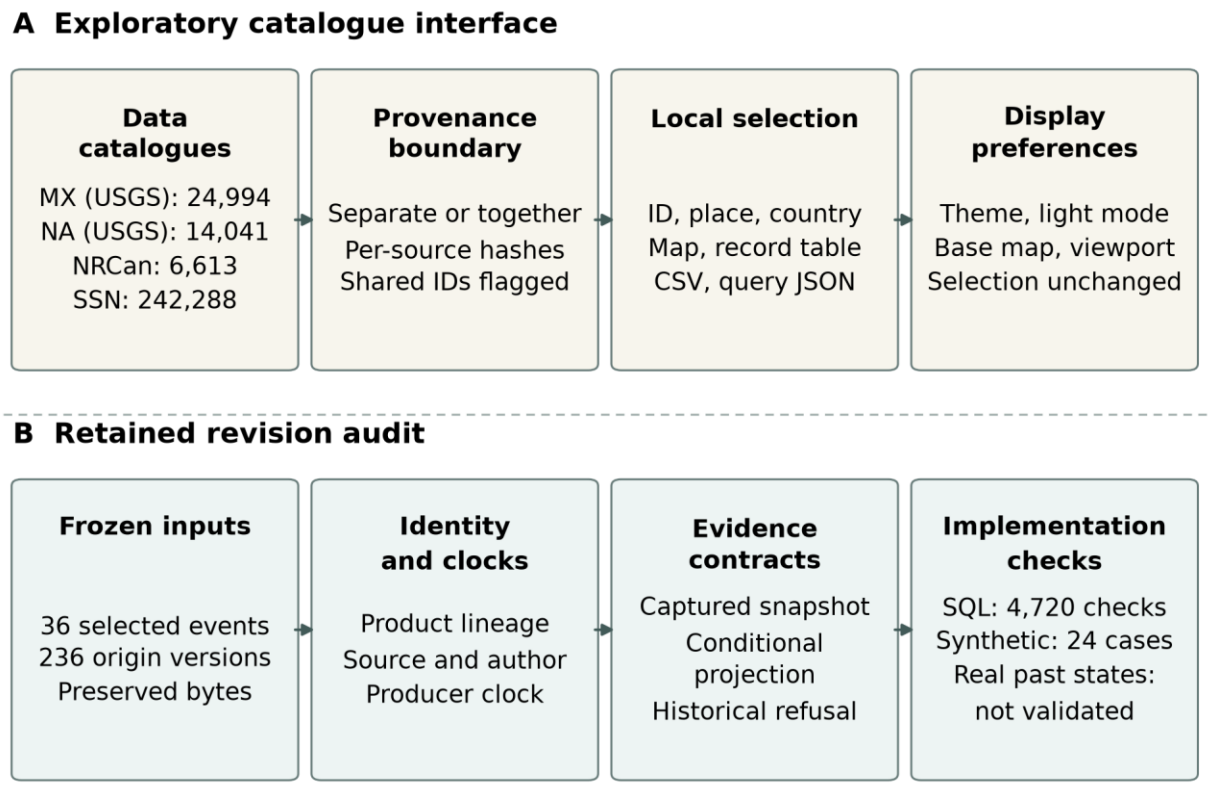


Figure 3. Architecture and evidence flow. Catalogue-specific CSV snapshots and manifests feed the browser explorer and its CSV/local-query JSON exports. Captured ComCat responses feed the conditional retained-lineage analysis and its independent SQL checks. Explicitly synthetic histories separately exercise specified evidence-loss mechanisms. Package hashes and source roles are retained across both workflows. The exploratory selections do not become additional revision-validation cohorts or reconstructed historical public states; map backgrounds and presentation preferences are display inputs. The Matplotlib diagram code was developed with OpenAI Codex assistance (OpenAI; GPT-6-based agent session, 6 November 2026).

Record inspection identifies source, key kind, fields and hashes. Figure 4 shows the Canadian case, preserving a local key and raw depth without converting them to an original event ID or documented kilometre measurement.

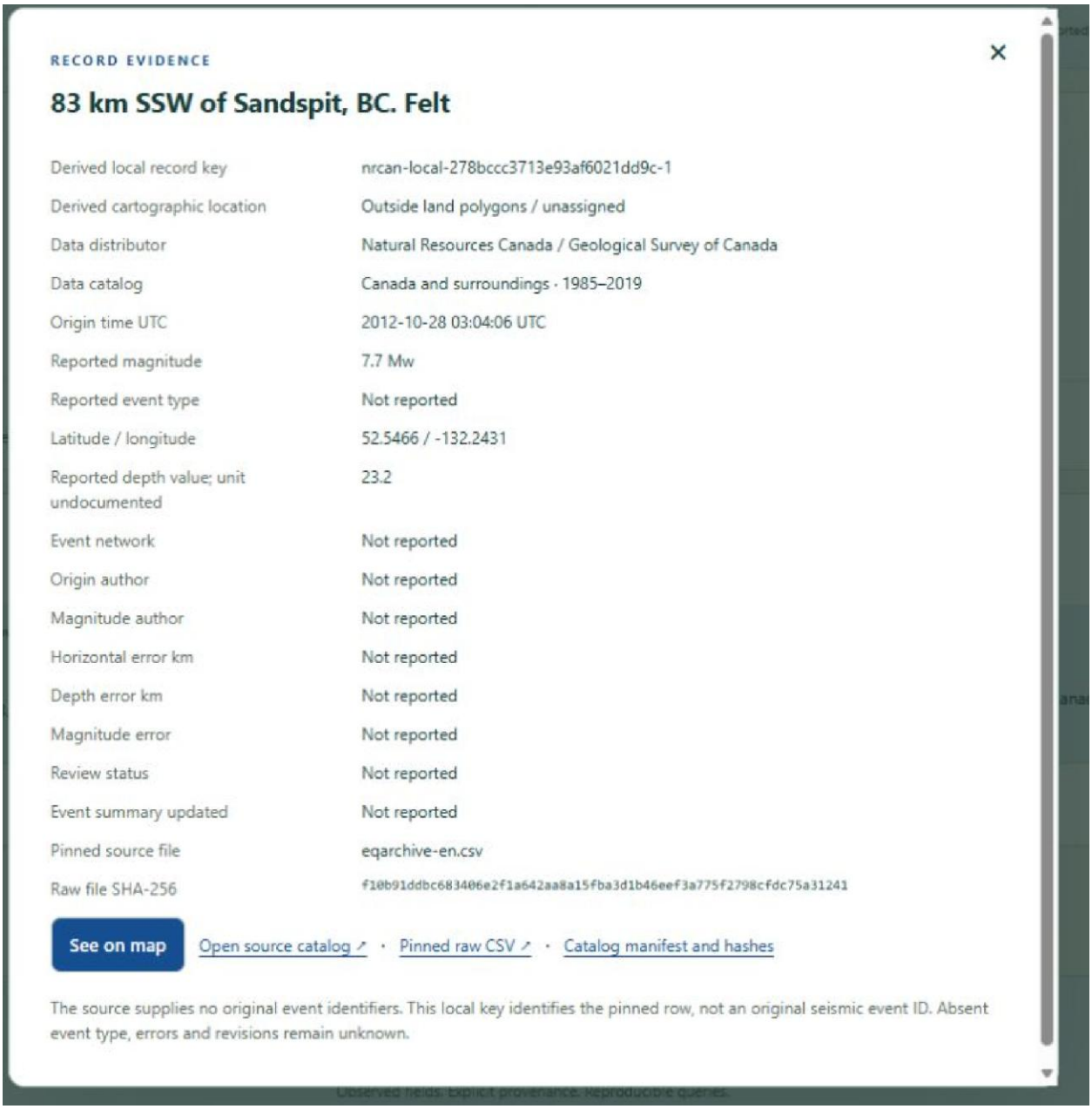


Figure 4. Canadian record inspection in the released system. The source-specific view exposes the catalogue, derived local identifier, reported fields and provenance links for the 28 October 2012 row reporting 7.7 Mw. Its raw depth value of 23.2 is retained without assigning a kilometre unit. The land-polygon label is derived separately from the distributor. The image frames one genuine screen capture, without combining separate states. This capture demonstrates the inspection workflow, not an independent revision validation or a comparison with a USGS observation of the same earthquake.

Implementation and figure code were prepared with assistance from OpenAI Codex (OpenAI; GPT-6-based agent session, 6 November 2026). The maintained Python pipelines render Figure 3 and Figures 5-8 with Matplotlib 3.11.2 from declared reports and records. Figures 1, 2 and 4 use genuine Microsoft Edge captures of the released system served locally, with each view state and image hash retained. The captures contain no generated or retouched observations.

2.2. Product identity and source roles

Product lineage is (source, type, code); updateTime identifies its version. Exact four-part repeats are deduplicated; conflicting content is rejected. The distributor remains distinct from origin-source, magnitude-source, eventsource and eventsourcecode, following PDL's product/event distinction. Missing values remain unknown.

Primary endpoint analysis requires source us, type origin and code equal to the canonical event identifier at capture. Filtering source alone would admit origins authored by another agency. Public origins remain separate from internal-origin and phase-data. Current identifiers and associations are not backdated.

2.3. Executable evidence contracts

The standard-library module implements three outputs (Table 1). The first queries captured summary bytes, retaining their hash and supplied receipt time; it neither authenticates that clock nor assumes an unchanged global catalogue during sequential retrieval.

The second selects each lineage's greatest producer updateTime not exceeding a cutoff, then filters attributes. Filtering first could retain an older above-threshold version after a newer version falls below the threshold. A selected DELETE is reported separately without restoring its predecessor. Multiple lineages remain multiple product records.

Table 1. Contract	Interpretation of the output	Principal boundary
snapshot_ observed	Query result on supplied current summary bytes at capture	Not a historical replay or a complete physical population
retained_ projection_ conditional	Latest retained version per lineage before a producer-time cutoff	Neither public availability nor historical preference is established

Table 1. Contract	Interpretation of the output	Principal boundary
historical_ catalog_ not_ identifiable	Stronger historical catalog claim is declined	Missing historical evidence is not manufactured

Projection requires explicit acknowledgement of retained-only evidence, producer/publication clock differences and unknown past associations/preferences. Its selected_lineage_count counts lineages. The offline projection module supports magnitude bounds, exact magnitude/event types, coordinate bounds and half-open origin-time intervals. Null attributes fail value-dependent predicates; magnitudes are not converted.

The third output declines historical public-state reconstruction because past membership, first availability, retention completeness, preferences, management records and associations are unestablished. This does not assert that particular missing versions existed.

2.4. Endpoint and selected-field comparisons

Endpoints are the earliest and latest retained non-deletion versions, ordered by producer time, not necessarily first release and reference truth. Magnitude differences require available values and exactly matching, case-preserved types. Crossings compare their membership at inclusive thresholds 5.5, 6.0, 6.5, 7.0, 7.5 and 8.0; within-type variation across all retained versions is also checked.

Depth differences use reported kilometres. Haversine epicentral displacement uses radius 6,371.0088 km; it measures separation, not hypocentral distance, location error or accuracy improvement. Decimal precision supplies no uncertainty interval.

Adjacent versions are compared over 27 declared fields: time, coordinates, depth/type, magnitude/type, agencies, uncertainties, gap, station distance, phase/station counts and error-ellipse attributes. Numeric equality and missing values are preserved. “Unchanged selected fields” does not establish equality of other properties, embedded contents or phase picks.

2.5. Counterexamples and documented comparators

Three synthetic paired histories share supplied current evidence but differ in past public release, past preference, or an earlier withdrawn event absent from enumeration. Independent symbolic replay verifies identical current evidence and different past answers, demonstrating logical possibility rather than prevalence in ComCat.

Tests cover hashes, nulls, selection-before-filtering, deletions, conflicts, ambiguous associations and distributor/author separation. They validate code, not clock accuracy or archive completeness.

Documentation and accessible source interfaces establish capabilities of libcomcat, Timeline and INGV x-fdsnws-event; no runtime, accuracy or usability benchmark is performed. Timeline's inspected GitHub implementation is historical; its maintained code.usgs.gov destination was inaccessible, so current equivalence is not assumed.

2.6. Independent evaluation and controlled evidence loss

SQLite independently normalizes original products into relations and ranks versions by updateTime within (source, type, code) before attribute filtering, instead of Python dictionary selection. Decimal parsing and integer UTC times provide an additional implementation path. Comparisons inspect identities, timestamps, source roles, measurements/types, deletions and pre-filter counts.

Each distinct retained origin update is evaluated immediately before and at its timestamp. Ten queries cover unfiltered results, magnitude thresholds, exact mww/mb/mB, the regional rectangle, half-open 2017/2022 intervals and earthquake type. Empty mB results test case preservation without adding an observed sample. Input lengths and hashes are verified.

Eight deterministic synthetic histories exercise omitted magnitude/spatial revisions, deletion, absent publication/preference, withdrawn membership, filtering before selection and index/producer clock substitution. A regional filter and three magnitude variants (unbounded, ≥ 5.5, ≥ 6.1) apply to each. Complete histories, exports and omissions remain explicitly synthetic fixtures.

Independent public-action replay supplies event counts. Three illustrative backcasting policies use current enumeration/preference with producer-time selection, filtering first, or index-time substitution; they are not existing software implementations. Their event counts are compared with replay, while conditional lineage results are compared with SQL. Historical refusal is prescribed conservative behavior, not a detector of specific missing records.

3. Seismic data sources and study design

3.1. Documented catalogues and their coverage

Four catalogues demonstrate source-aware exploration (Table 2), separately from revision validation. Their three distributors are USGS ComCat, Natural Resources Canada / Geological Survey of Canada (NRCan/GSC), and Servicio Sismológico Nacional / Instituto de Geofísica, UNAM (SSN). Origin-time coverage is not a sequence of public catalogue states. Distributors can carry another agency's solution, and the catalogues can share observations.

Table 2. Catalogue and distributor	Origin-time coverage	Records	Interpretation limits
Mexico-region diagnostic / USGS ComCat	1900-2026	24,994	Earthquake records; regional rectangle; two quarry blasts excluded; not an independent SSN catalogue
North American regional windows / USGS ComCat	2020-2025	14,041	Earthquake records from twelve pinned CSVs; regional windows, not national assignments or a revision archive
Canada and surroundings / NRCan/GSC	1985-2019	6,613	Reported $M \geq 3.5$; five repeats retained; original event IDs, event types and uncertainty/revision fields absent
Mexico and surroundings / direct SSN	1900-2026 (partial)	242,288	Reported $M \geq 3.5$ selected from 402,082 source rows; magnitude types, original IDs and revision fields absent; coverage is not completeness

The North American input combines latitude 14-84°, longitude -180 to -50°, with an Aleutian window at latitude 50-73°, longitude 170-180°. Its upstream manifest reports extraction on 26 October 2026. The 770 positive-longitude records retain their original coordinates; longitude wrapping affects display only. These windows include offshore and neighboring areas.

The official Canadian CSV was acquired on 6 November 2026 under the Open Government Licence-Canada (Natural Resources Canada, n.d.; Government of Canada, n.d.). Its 101,365 source rows yield 6,613 records after excluding 94,752 values below 3.5. Exact repeats are preserved as separate records, not asserted to be distinct physical earthquakes. All reported magnitude-type strings retain their case. Source dates contain an explicit UTC offset. Where original event IDs are absent, content-derived local row keys are marked as such. The particular CSV does not document a depth unit: its literal depth value is preserved, while the common kilometre field remains null. This licence applies to the linked Open Government resource, not automatically to the separate NEDB bulletin.

The official SSN download on 6 November 2026 requested 1900-2026-11-06 in UTC, all magnitudes, depths and states (Servicio Sismológico Nacional, 2026a). Its ten columns preserve UTC and local date/time, magnitude, coordinates, depth in documented kilometres, reference locality and literal status. The 402,082 rows comprise 384,433 numeric magnitudes and 17,649 no calculable values; the browser selects 242,288 with reported $M \geq 3.5$. Original event IDs, magnitude/event types, uncertainty estimates, update times and revision histories are absent. Local content-derived keys identify rows; missing fields remain unknown. verificado and revisado are source statuses, not inferred revision sequences. The input hash fixes this download, whose 2026 coverage is partial. Its generation label, 13:57:07 central Mexico time, nominally corresponds to 19:57:07 UTC; the latest reported origin is 20:31:07 UTC. This discrepancy precludes treating the label as an atomic capture clock. Eight UTC/local pairs have atypical offsets; neither column is corrected. UTC analysis uses the explicitly labelled source columns.

Descriptive annual bins span every calendar year from 1900 to 2026. Years without source rows remain gaps; annual maxima are one scalar per observed year, with all tied rows retained as provenance. Reference localities do not assign territorial or tectonic regions. Magnitudes are not homogenized.

The SSN data-specific free-use terms permit use with source acknowledgement, preserved meaning and third-party rights, without institutional endorsement or unauthorized logos (Servicio Sismológico Nacional, 2026b). The original CSV's non-commercial reproduction notice is retained alongside those terms; no Creative Commons licence is inferred. Historical completeness is not transferred from another study: the public abstract of Zúñiga et al. (2000) identifies a reporting change in 1988 and M_d 4.3 completeness for 1988-1998, not a universal cutoff for this download.

3.2. A separate provenance diagnostic

A pinned upstream snapshot, commit 8fe055ffe635875f7290906d45f70562a7896621, supplies a separate CSV provenance diagnostic from the Actronix09 fuentes-alternas repository (https://github.com/Actronix09/fuentes-alternas/tree/8fe055ffe635875f7290906d45f70562a7896621/sismos-usgs). Its 46 files contain 24,996 unique identifiers: 24,994 earthquake records and two quarry blasts. Among the earthquake records, 9,749 have an exact unm value in locationSource or magSource; including the composite value us_unm gives 9,858 records. The official contributor directory (https://earthquake.usgs.gov/data/comcat/contributor/unm/) identifies UNM with the Instituto de Geofísica, UNAM. This demonstrates that a record accessed through USGS infrastructure cannot automatically be classified as an independent USGS determination. The EMSC contributor directory identifies UNM as the Servicio Sismológico Nacional at the Instituto de Geofísica, UNAM (European-Mediterranean Seismological Centre, n.d.). The explorer therefore labels its contributor access as SSN/UNAM via ComCat, preserving USGS as the distributor and the source codes.

3.3. A bounded cohort for revision analysis

The empirical pilot comprises two current-catalog queries for calendar years 2017 and 2022. Both require event type earthquake, preferred magnitude at least 5.5, latitude from 14 to 33 degrees and longitude from -118 to -86 degrees. This numerical selection uses reported preferred magnitudes and does not define a magnitude-homogeneous catalog. The rectangle includes offshore areas and neighboring countries; it is not a territorial classification of Mexican earthquakes. The year-specific origin-time intervals extend from 1 January at 00:00:00.000 UTC to 31 December at 23:59:59.999 UTC.

Acquisition started at 19:53:15.316 UTC and finished at 19:53:54.609 UTC on 5 November 2026. The count endpoint was checked for each cohort before retrieval. The workflow rejects a response exceeding the service maximum rather than accepting truncation. Query results must agree with the acquired count and contain unique event identifiers. For each returned identifier, a detail request uses format=geojson and includesuperseded=true. Original response bodies are saved without reserialization, alongside request and response URLs, retrieval start and completion times, response metadata, byte lengths and SHA-256 hashes.

Selection uses preferred parameters available during acquisition. Events that previously satisfied the filters but subsequently moved below the threshold, outside the rectangle, to another year, or out of the current catalog are not systematically recovered. Consequently, the cohort supports an audit of these retrieved events, not an estimate of all

historical threshold crossings or withdrawals. The two years are demonstration cohorts, not a representative sample of catalog maintenance behavior.

4. Results from the system and reproducible analyses

4.1. From source selection to reusable information

The demonstrations address the user's progression from selecting evidence to obtaining information for further investigation. The default North American view exposes 14,041 records; the Canadian catalogue supplies 6,613 historical records with its own coverage and missing fields (Table 2). A country/territory filter can be applied within either a single catalogue or the combined view; it selects land-polygon containment independently of the reporting agency and keeps outside/boundary categories explicit.

For a recent-record study, selecting direct SSN, UTC origin year 2026 and reported $M \geq 4.5$ returns 204 records through 6 November (Figure 1; partial year). This is a reproducible subset for inspecting locations and reported parameters, not all Mexican seismicity. Native SSN supplies 242,288 packaged records, whereas the separate SSN/UNAM-via-USGS shortcut selects 9,858 Mexico-package rows, including 109 admitted by the composite us_unm value. The labels prevent mistaking that older contribution filter for the direct SSN download.

For joint exploration, selecting all four catalogues, UTC years 2010-2025 and reported $M \geq 4.5$ gives 7,107 source records (Figure 2). Contributions remain visible: 2,114 North American USGS, 2,026 Mexico-region USGS, 377 NRCan and 2,590 SSN records. The expanded map displays 7,073 within its frame; all 7,107 remain in the selection and export. This distinguishes a query result from viewport visibility, while preserving different source coverage and shared observations. Counts do not become a combined estimate of unique earthquakes.

Record inspection then exposes the evidence behind a selected marker. The Canadian example retains its local key, unknown event type and raw depth without inventing a kilometre unit (Figure 4). Its link leads to the Canadian catalogue rather than a USGS page constructed from that local key. CSV retains selected records and provenance; query JSON retains input hashes, predicates and ordered identifiers. These outputs provide an explicit starting dataset for external analyses, with derived geography distinguished from source fields. Browser checks verify catalogue switching, altered-package rejection, source links and shared views; they do not measure user effectiveness or source accuracy.

The following results show what additional descriptive information can be derived from preserved inputs using the accompanying offline pipelines. The revision pilot examines parameter changes and their consequences for

selection; the SSN summaries examine observed coverage and annual maxima. These analyses remain separate from the browser's interactive record counts.

4.2. Derived source attribution in the revision pilot

The acquired cohorts contain 22 events for 2017 and 14 for 2022. Their detail responses contain 236 public origin-product versions, all with status UPDATE. All 36 captured canonical identifiers have a primary US origin lineage with multiple retained non-deletion versions; these lineages contain 205 versions. Superseded versions retain status UPDATE, so this count does not imply 205 simultaneously active solutions. Absence of retained deletion products in this selected cohort is not evidence that events were never withdrawn or that all versions have survived.

Table 3. Observed quantity	Value	Counting unit
Current-selected event identifiers	36	Captured events
Public origin-product versions	236	Retained versions
Versions in primary US origin lineages	205	Retained versions
Distributor source us	226	Public origin versions
Distributor source at / pt	6 / 4	Public origin versions
Origin author us / iscgem / missing	205 / 21 / 10	Public origin versions
Separate internal-origin / phase-data versions	15 / 203	Related products, excluded from primary analysis

The difference between 226 US-distributed versions and 205 US-authored versions is explained here by 21 origins carrying iscgem authorship. The remaining ten public origins have missing solution-agency attributes in the fields

inspected. No agency identity is imputed for them. These observed field relationships illustrate the need to retain source roles separately rather than treating each distributor code as an independent observational source.

4.3. Derived parameter changes and threshold membership

Across the 36 endpoint pairs, median epicentral displacement is 12.339 km and maximum displacement is 79.684 km. The latter belongs to us2000ahv7, whose endpoint magnitudes both remain 5.7 mb. Depth differences range from -38.48 to 12.39 km. Neither displacement nor depth change establishes which solution is closer to the physical source (Figure 5).

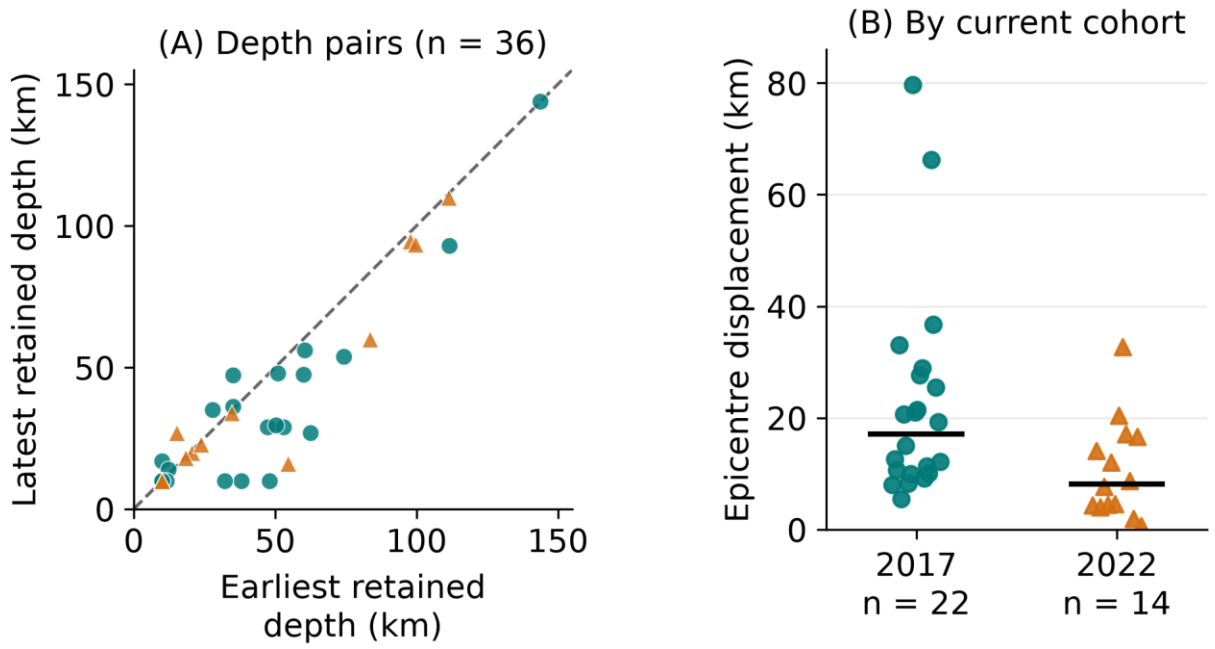


Figure 5. Depth and epicentral differences between earliest and latest retained non-deletion versions within primary US origin lineages. (A) Paired depths for 36 events; dashed line indicates equal endpoints. Circles identify 2017 events and triangles identify 2022 events. (B) Epicentral displacement by current cohort, comprising 22 events in 2017 and 14 in 2022. Horizontal offsets expose overlapping observations; black bars mark cohort medians. Differences do not estimate error relative to independent source locations.

Thirty-one events have comparable endpoint magnitudes with unchanged types: 30 mww pairs and one mb pair (Figure 6). Thirteen have a nonzero difference; the minimum, median and maximum differences are -0.1, 0.0 and 0.2 magnitude units. Five endpoint pairs have different reported types and are excluded from that subtraction. Same-

type comparison controls one source of incompatibility but does not establish constant measurement procedures or remove selection bias.

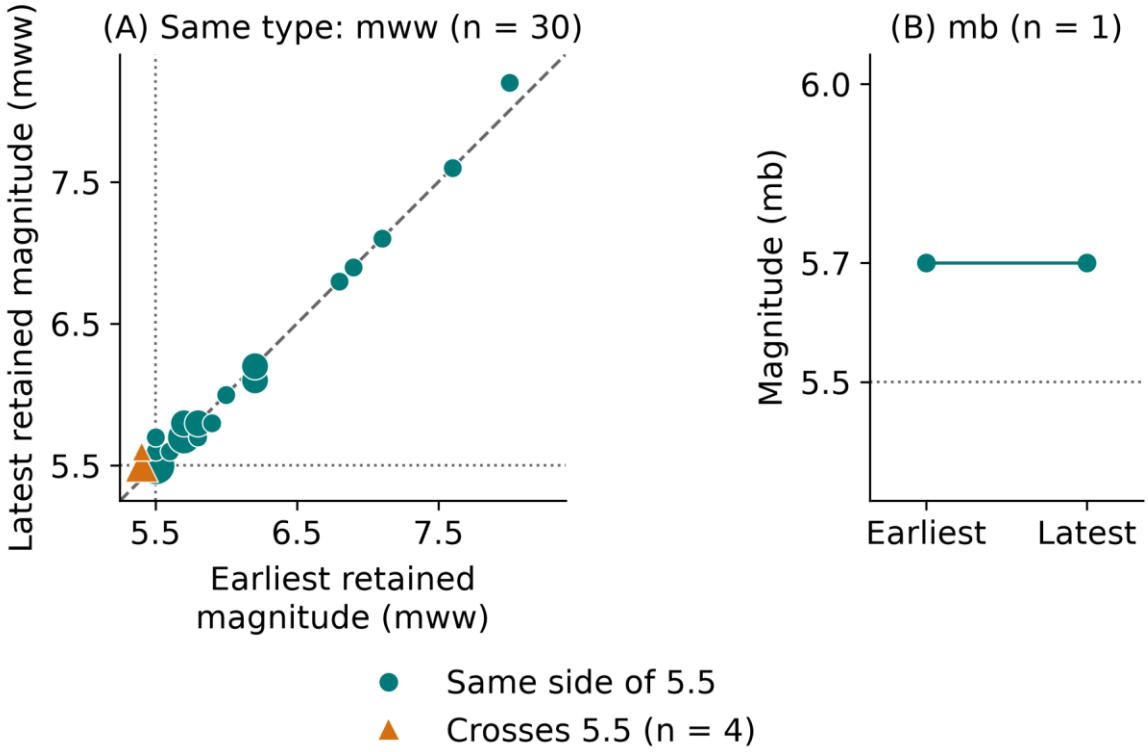


Figure 6. Magnitude endpoint comparisons restricted to exact matching types. (A) Thirty mww pairs; marker area represents multiplicity of identical pairs. Orange triangles represent four events crossing 5.5. Dotted lines indicate 5.5; dashed line indicates equal endpoints. (B) One mb pair, displayed separately. No scale conversion or direct cross-scale comparison is performed. Current selection at magnitude at least 5.5 prevents interpreting these crossings as an unbiased estimate of threshold migration.

Four endpoint pairs cross the 5.5 threshold, all within mww series in the 2017 cohort (Table 4). No comparable endpoint pair crosses the other examined thresholds. The within-type retained-series check identifies the same four events at 5.5. Because current magnitude at least 5.5 defined admission, the observed upward crossings cannot estimate a general rate or direction of catalog threshold migration.

Table 4. Event identifier	Earliest retained mww	Latest retained mww
us2000ailq	5.4	5.5

Table 4. Event identifier	Earliest retained mww	Latest retained mww
us2000ap3b	5.4	5.6
us2000aqji	5.4	5.5
us1000arzf	5.4	5.5

The primary lineages contain 169 adjacent version pairs. Selected fields differ in 117 pairs and remain unchanged in 52. These counts depend on the declared 27-field representation. They should not be interpreted as numbers of analyst actions, substantive revisions or redundant complete products.

4.4. Clock distinctions and conditional historical claims

For us2000ahv0, the seven retained primary US origins span endpoint values from magnitude 8.0 to 8.2 mww and depth 35.0 to 47.39 km. The latest retained primary origin update is dated 7 December 2017. A separate retained origin branch has an update in May 2022, while the captured event-summary update is dated 8 October 2026. Inspection of the captured products associates that latest summary timestamp with DYFI. It is consequently not the date of the primary hypocenter's latest retained revision. Figure 7 displays only retained versions within the primary US origin lineage, without combining other origin families or non-origin updates.

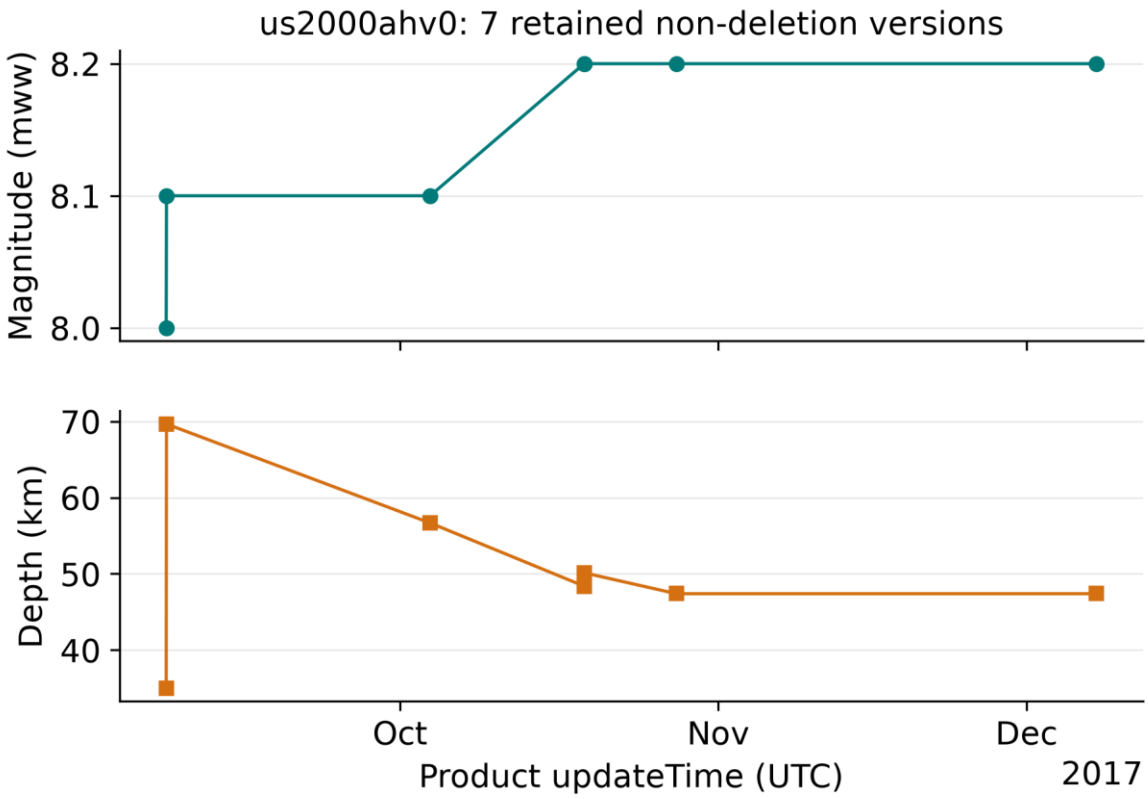


Figure 7. Reported mww magnitude and depth in seven retained non-deletion versions of product lineage (us, origin, us2000ahv0), ordered by producer updateTime in UTC. Connecting lines aid reading; they do not reconstruct continuous source evolution, public availability or historical preference. Earliest retained version is not assumed to be first issued.

Ten of the 36 event summaries have an update timestamp later than every retained public origin timestamp. A summary timestamp may therefore reflect activity outside the origin products examined. Even where timestamps coincide, their equality alone does not establish publication latency or the historical preferred-origin sequence. Product update, indexing, retrieval and physical origin time remain different clocks.

The executable contracts preserve these distinctions. Captured summaries support capture-scoped counts, and supplied retained products permit explicitly conditional lineage projections. Neither operation supplies the missing evidence required for an unconditional past public catalog. The three synthetic counterexamples pass the replay checks while maintaining different past answers. Their logical result complements the observational audit; it is not an empirical estimate of how often a historical reconstruction would fail.

4.5. Differential validation and constructed evidence loss

The 36 captured details contribute 236 distinct event-specific origin update boundaries. Evaluating both sides gives 472 producer-time cutoffs and 4,720 query evaluations across the ten specifications. Selected records and declared deletions agree with the independent SQL implementation in every evaluation. This checks retained-projection implementation on the acquired inputs; it does not validate historical public availability, completeness or source accuracy. Real origin products all have status UPDATE, so their deletion comparisons are empty; deletion handling is exercised with synthetic inputs.

The eight constructed histories produce 24 synthetic query cases. Conditional retained projections agree with SQL in all 24 cases. Relative to known public event counts, producer-time backcasting with current preference differs in ten cases, filtering before temporal resolution in eleven, and index-time backcasting in ten. These policies share current-cohort and current-preference assumptions, so their failures can overlap and are not isolated causal effects of a single operation. The examples were constructed to exercise specified failure mechanisms and controls; the counts do not estimate operational error prevalence or establish superiority over existing catalog software.

All 24 unconditional historical claims are declined as designed. This outcome must not be interpreted as perfect loss detection: the contract declines the stronger claim regardless of whether a particular synthetic export omitted a version. Its conditional calculation remains reproducible, while the known public replay can differ because it uses additional retained-history, publication, membership or preference evidence.

4.6. Observed SSN coverage and annual summaries

Figure 8 distinguishes downloaded-record coverage from physical occurrence. Of 126 closed calendar bins in 1900-2025, 110 contain records and 16 contain none in this download. All 110 observed years retain at least one $M \geq 5$ row: that threshold removes no additional national years. Across the full download, 1,901 rows have $M \geq 5$. The partial 2026 bin contains 34,548 source rows, 14,875 with $M \geq 3.5$ and 35 with $M \geq 5$. These changing counts reflect reporting and selection as well as seismic activity; they establish neither completeness nor changing hazard. The 110 closed-year maxima are attained by 118 rows because eight years have ties. Counting those rows as 118 independent annual maxima would change the statistical unit. Unknown magnitude types, uneven observation coverage and unverified stationarity prevent interpreting fitted tails as validated return periods. Accordingly, no occurrence forecast, regional independence claim or design magnitude is derived from these descriptive summaries.

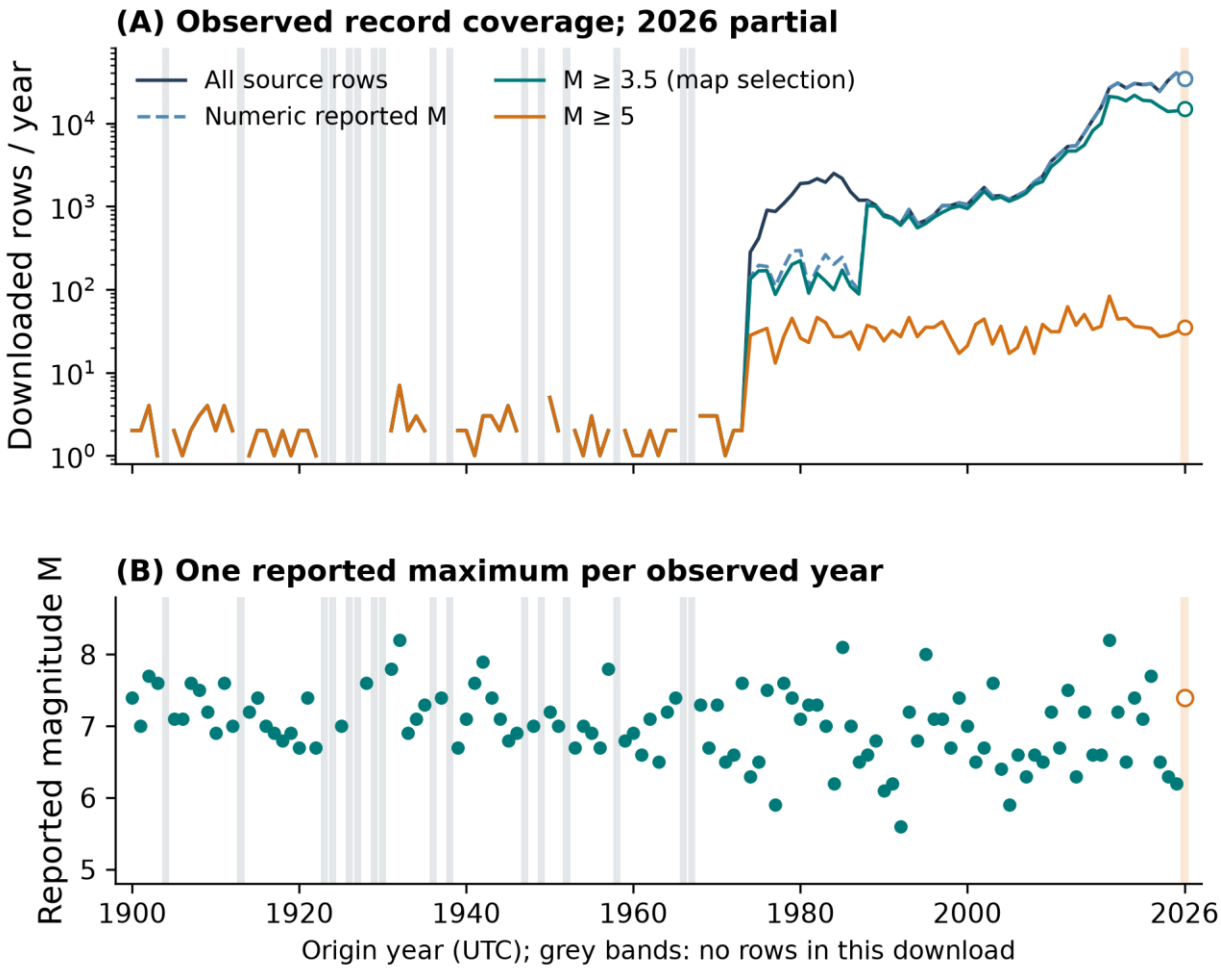


Figure 8. Descriptive coverage of the preserved SSN download. (A) Annual row counts on a logarithmic axis, separating all source rows, numeric magnitudes, the M ≥ 3.5 cartographic selection and M ≥ 5. (B) One maximum reported magnitude per observed year. Grey bands mark the 16 closed-year bins without source rows; these gaps are not physical zero counts. Open circles and the orange band identify partial 2026. Lines aid reading of downloaded counts; neither panel establishes completeness, homogenized magnitude, hazard or a return-period forecast.

5. Discussion

The system addresses the stated problem by connecting exploration to evidence that remains inspectable after export. Coordinated filters turn heterogeneous catalogues into explicit study selections; source-qualified records and hashes make those selections reusable. The demonstrations show how a researcher can isolate recent SSN records, inspect a Canadian entry and examine several sources together while retaining their differences. Descriptive outputs

support the formulation of research questions and preparation of subsequent analyses. They do not remove the need for study-specific completeness assessment, magnitude harmonization or event reconciliation.

The derived results also show why provenance affects interpretation. SSN annual counts and maxima expose coverage gaps and tied observations that matter when defining an annual analysis. The revision pilot shows how parameter changes affect a magnitude threshold or spatial comparison. Separating the distributor from the solution agency exposes ISC-GEM origins carried by USGS; separating summary updates from origin revisions avoids interpreting unrelated product activity as a hypocentral change. Evidence contracts attach these boundaries to the computation rather than leaving them implicit in a map.

This functionality builds on existing software rather than replacing its capabilities. Libcomcat documents product-history extraction; Timeline groups versions and presents their evolution. The INGV service resolves event versions before filtering and provides asofdate under an invariant that all public mutations pass through its API. That controlled maintenance boundary differs from a researcher retrieving evidence from another provider. The present prototype specifies which weaker calculations remain permissible when the stronger boundary cannot be established. This is a comparison of documented semantics, not proof of superior performance or a comprehensive absence of equivalent methods.

Parameter differences can alter spatial-window membership or threshold counts, but latest retained solutions are not reference truth. The SSN and Canadian integrations expose heterogeneous fields without providing additional revision validation. Local keys do not restore original event IDs; magnitude filtering does not establish scale equivalence. Catalogue origin-time coverage is not an archive of past public states. Additional controls and map backgrounds add no independent seismic evidence.

Retrospective selection omits events outside the current cohort, including possible downward crossings and withdrawals. Unknown retention, past associations and preferences prevent historical reconstruction; synthetic experiments cannot replace independently captured public states.

The demonstrations do not evaluate independent users, browser performance or research outcomes. Reproducible selections establish functional behavior, not measured improvements in usability or scientific productivity.

Further evaluation requires independent historical captures, additional cohorts and prespecified comparisons. SQL agreement and constructed-loss experiments establish neither historical accuracy nor operational error prevalence or transferability.

6. Conclusions

The developed web system supports consultation, comparison and exploration of heterogeneous earthquake catalogues while retaining the provenance required to interpret and reuse results. Its three connected outputs are the computational application, documented seismic inputs and traceable derived information. Source-specific and combined views, record inspection and hash-linked exports demonstrate how users can turn dispersed records into explicit selections for analysis, study and subsequent research.

The accompanying reproducible pipelines demonstrate further uses of those inputs: descriptive SSN coverage and annual maxima, source attribution and retained-parameter comparisons. In the 36-event pilot, magnitude changes, threshold crossings and epicentral differences provide concrete reasons to preserve parameter versions and source roles. SQL agreement validates the tested conditional computations; it does not establish historical public states or source accuracy. The system therefore supplies reusable evidence and bounded analytical results, with uncertainty and missing information visible. Independent user evaluation, broader scientific cohorts and independently preserved historical states are needed to evaluate its utility and generalizability beyond the demonstrations presented here.

Code and data availability

Released code, source manifests, the frozen ComCat evidence package and reproducible validation are available in the study repository (https://github.com/gabrielhuav/Seismic_Data_Geosciences). The geographic/composition extension is released in code commit 49372d02198a614a07dda1a3ecbb8a21d754ff68. Published application assets match those retained for the local browser captures. The package retains initial-release revision/Mexico-summary bytes and catalogue-specific provenance: 36 event-detail responses and four cohort count/query responses with receipts and SHA-256 hashes. Run python scripts/prepare_public_evidence.py restore before python scripts/audit_revisions.py; offline restoration rejects different existing evidence. Live URLs cannot substitute for preserved bytes.

The separate CSV diagnostic is pinned by commit and hashes. Obtain missing files with python scripts/audit_sources.py --download --skip-ssn. Historical SSN-derived SQL is not redistributed or required. Source rights and contributor roles remain documented; MIT covers original software, not observations or third-party material. No archival dataset DOI is asserted.

The Canadian source CSV and its manifest are preserved in research-data/catalogs/nrcan-ca/ under the Open Government Licence-Canada; python scripts/build_canada_catalog.py reproduces its browser package offline. python scripts/build_catalogs.py --download obtains and verifies missing North American CSVs at the pinned upstream commit. The registry identifies each package separately, including hashes and provider-specific limits. The preserved native SSN CSV is research-data/catalogs/ssn-mx/ssn_19000101_20261106.csv, SHA-256 0601c65544bc95ef865d3440e5e6d84249acdebd8edfb379669597096543f993. python scripts/build_ssn_catalog.py reproduces its selection offline. The source catalogue DOI identifies the mutable SSN service, not an archival DOI for this download. Its original notice and observed data-use terms are retained; SSN, Instituto de Geofísica, Universidad Nacional Autónoma de México, is credited for the data. No institutional endorsement is implied. Run python -m unittest discover -s tests for contract checks and python -m scripts.validate_evidence for the independent SQLite and synthetic controlled-loss evaluation. Generated manifests and tables distinguish original observed details from synthetic histories. The published browser explorer (https://gabrielhuav.github.io/Seismic_Data_Geosciences/) exposes source-file evidence and exports local-query specifications with input hashes and result identifiers; its map is not an operational prediction or alert service.

CRediT authorship contribution statement

José Manuel Villa Vargas: Conceptualization; Formal analysis; Supervision; Writing - original draft; Writing - review & editing.

Gabriel Hurtado Avilés: Data curation; Investigation; Methodology; Project administration; Resources; Software; Validation; Visualization.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare no competing interests.

Declaration of generative AI assistance

OpenAI Codex (OpenAI; GPT-6-based agent sessions, 6-7 November 2026) assisted literature discovery, code development, reproducible figure scripts, automated testing, manuscript drafting and document formatting. The

explanatory diagram and data plots are rendered from documented inputs and maintained code. The interface panels derive from genuine browser captures, without generative image creation or retouching.

References

European-Mediterranean Seismological Centre, n.d. Data contributors. Accessed 6 November 2026. Contributor directory (https://emsc-csem.org/Earthquake_data/contributors.php).

Gardezi, J., Bertossi, L., 2011. Query Answering under Matching Dependencies for Data Cleaning: Complexity and Algorithms. Author preprint, arXiv:1112.5908. Preprint (https://arxiv.org/abs/1112.5908).

Government of Canada, n.d. Open Government Licence-Canada, version 2.0. Accessed 6 November 2026. Licence (https://open.canada.ca/en/open-government-licence-canada).

Natural Earth, n.d. About Natural Earth. Geographic data and boundary policy. Accessed 6 November 2026. Project documentation (https://www.naturalearthdata.com/about/).

Natural Resources Canada, n.d. Earthquakes in Canada. Open Government Portal, record 4cedd37e-0023-41fe-8eff-bea45385e469. Accessed 6 November 2026. Official dataset (https://open.canada.ca/data/en/dataset/4cedd37e-0023-41fe-8eff-bea45385e469).

Pintore, S., 2026. x-fdsnws-event: a reference implementation of a version-aware extension of the FDSNWS/event access model for reproducible seismic event catalogues. INGV software repository. Accessed 5 November 2026. Repository (https://github.com/INGV/x-fdsnws-event).

Sawires, R., Santoyo, M.A., Peláez, J.A., Corona Fernández, R.D., 2019. An updated and unified earthquake catalog from 1787 to 2018 for seismic hazard assessment studies in Mexico. Scientific Data 6, 241. Article (https://doi.org/10.1038/s41597-019-0234-z).

Servicio Sismológico Nacional, 2026a. Catálogo de sismos, México. Instituto de Geofísica, Universidad Nacional Autónoma de México. Downloaded 6 November 2026. Catalogue DOI (https://doi.org/10.21766/SSNMX/EC/MX).

Servicio Sismológico Nacional, 2026b. Aviso legal, uso de datos sísmicos y términos de libre uso. Observed 6 November 2026. Terms (http://www.ssn.unam.mx/aviso-legal/).

Sismanis, Y., Wang, L., Fuxman, A., Haas, P.J., Reinwald, B., 2009. Resolution-Aware Query Answering for Business Intelligence. Proceedings of the 25th IEEE International Conference on Data Engineering, 976-987. Publication (https://research.ibm.com/publications/resolution-aware-query-answering-for-business-intelligence).

Thompson, E.M., McBride, S.K., Hayes, G.P., Allstadt, K.E., Wald, L.A., Wald, D.J., Knudsen, K.L., Worden, C.B., Marano, K.D., Jibson, R.W., Grant, A.R.R., 2020. USGS near-real-time products; and their use; for the 2018 Anchorage earthquake. Seismological Research Letters 91(1), 94-113. Article (https://doi.org/10.1785/0220190207). Official publication record (https://www.usgs.gov/publications/usgs-near-real-time-products-and-their-use-2018-anchorage-earthquake).

U.S. Geological Survey, n.d.a. API Documentation; Earthquake Catalog. Accessed 5 November 2026. Documentation (https://earthquake.usgs.gov/fdsnws/event/1/).

U.S. Geological Survey, n.d.b. How Products Work in ComCat. Accessed 5 November 2026. Documentation (https://earthquake.usgs.gov/data/catalog/products.php).

U.S. Geological Survey, n.d.c. Product Distribution Layer; Indexer Processing. Accessed 5 November 2026. Documentation (https://usgs.github.io/pdl/indexer.html).

U.S. Geological Survey, n.d.d. libcomcat Application Program Interface. Accessed 5 November 2026. Documentation (https://github.com/usgs/libcomcat/blob/master/docs/api.md).

U.S. Geological Survey, n.d.e. ComCat Timeline. Historical source implementation and current developer-resource directory. Accessed 5 November 2026. Inspected implementation (https://github.com/jmfee-usgs/comcat-timeline/blob/master/src/Catalog.ts); Official directory (https://earthquake.usgs.gov/data/software/).

Villa Vargas, J.M., Hurtado Avilés, G., Climent Hernández, J.A., 2026. Cuando México tiembla: la historia contada por los datos. AZCATL Revista de Divulgación en Ciencias, Ingeniería e Innovación 4(6), 28-33. DOI: 10.24275/AZC2026E1004. Published antecedent (https://azcatl.azc.uam.mx/index.php/azcatl/article/view/75).

Weatherill, G.A., Pagani, M., Garcia, J., 2016. Exploring earthquake databases for the creation of magnitude-homogeneous catalogues: tools for application on a regional and global scale. Geophysical Journal International 206, 1652-1676. Article (https://doi.org/10.1093/gji/ggw232).

Wu, X., Poorthuis, A., Zurita-Milla, R., Kraak, M.-J., 2020. An interactive web-based geovisual analytics platform for co-clustering spatio-temporal data. Computers & Geosciences 137, 104420. Article (https://doi.org/10.1016/j.cageo.2020.104420).

Zúñiga, F.R., Reyes, M.A., Valdés, C., 2000. A general overview of the catalog of recent seismicity compiled by the Mexican Seismological Survey. Geofísica Internacional 39(2), 161-170. Public abstract and metadata consulted. Official record (https://repositorio.unam.mx/contenidos/4133493).